

QI26-20

A Reproducible QUBO–QAOA Benchmark for Generalized Bipartite Assignment on NISQ Hardware

Verified progress · real-hardware evidence · preliminary novelty assessment

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PHASE 1 · EXECUTABLE ARTIFACT



What the Project Now Demonstrates

Five claims supported by the executed artifact and the archived IBM Quantum job.

1

GM-QAOA works – ideally

Feasible probability = 1.0 and state-preparation / full-circuit fidelity = 1.0 on all three fixed instances.

2

Standard QAOA is functional

It recovers optimal scores in multiple settings; on the 4×2 capacity instance, feasible mass rises to 43.99% at p=3.

3

The pipeline reaches a QPU

Both p=1 circuits ran in one ibm_marrakesh Runtime job and sampled the exact optimum (score 3.25).

4

Rigor and traceability are built in

Same instances, score, and decoder; exact and Hungarian ground truth; exported CSV/JSON; fixed seeds; archived job ID; claims checklist and compliance matrix.

5

A preliminary originality result

Within the searched corpus, no prior publication was found that combines exact-capacity assignment, standard vs. GM-QAOA, controlled same-job hardware execution, full-distribution errors, and transpilation-resource analysis.

THE DEFENSIBLE MESSAGE

There is real, measurable progress: the algorithms function, the workflow is reproducible, and the experiment reaches present-day quantum hardware.



Generalized Bipartite Assignment as QUBO

One mathematical model, three controlled instances, one common scoring and validation layer.

DECISION VARIABLES

$x_{ij} \in \{0,1\}$ · agent $i \rightarrow$ resource j

CONSTRAINTS

- Each agent is assigned exactly once.
- Resource use is either one-to-one or fixed by capacity vector C .
- All classical and quantum workflows use the same decoder.

IMPLEMENTED QUBO

$$Q(x) = - \sum_{ij} S_{ij} x_{ij} + \lambda_A \sum_i \left(\sum_j x_{ij} - 1 \right)^2 + \lambda_R \sum_j \left(\sum_i x_{ij} - C_j \right)^2$$

Penalty coefficients are generated programmatically; the QUBO-to-Ising mapping is numerically validated.

reference_3×3

9 qubits · 6 feasible

One-to-one assignment · optimum score 2.70

one_to_one_4×4

16 qubits · 24 feasible

Reproducible seed-2026 instance · optimum score 3.08

exact_capacity_4×2

8 qubits · 6 feasible

Capacity vector $C=(2,2)$ · optimum score 3.25



Seven-Stage Reproducible Research Pipeline

The notebook is an executable artifact, not a collection of disconnected code cells.



CONTROLLED COMPARISON

Same instances + same score + same feasibility decoder + same backend protocol

Ground truth

Exact optimum and classical gaps

Algorithm quality

Feasible / optimal probabilities

Hardware quality

Counts, TVD, Hellinger, fidelity

Physical cost

Depth, instructions, native CZ



Strong Classical Reference Points Were Established

The quantum results are interpreted against exact optima—not weak or ambiguous baselines.

3×3 one-to-one

2.70

GLOBAL OPTIMUM

Exact · Hungarian · Greedy · SA
all recover the optimum.

4×4 one-to-one

3.08

GLOBAL OPTIMUM

Exact and Hungarian are optimal.
Greedy = 2.80 · SA = 3.01.

4×2 exact capacity

3.25

GLOBAL OPTIMUM

Exact · Greedy · SA
all recover the optimum.

SCIENTIFIC ROLE OF THE BASELINES

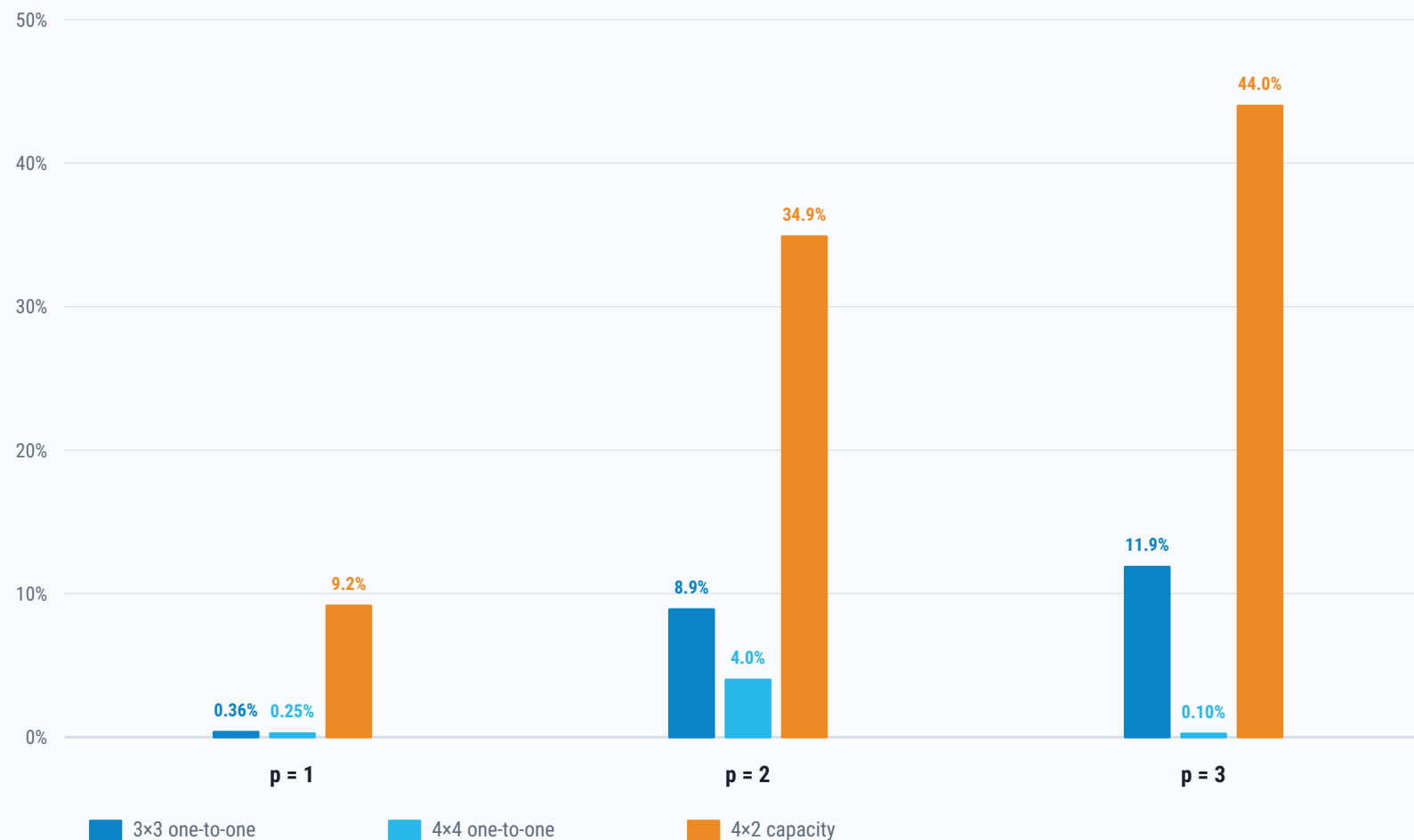
They establish the correct optimum, expose nontrivial heuristic failure on 4×4, and make zero-gap quantum samples interpretable.

Scope boundary: at these sizes, exact and Hungarian solvers are much faster. This is a correctness and NISQ implementation benchmark—not a runtime competition.



Standard QAOA Is Functional in Ideal Simulation

Feasible probability by depth p ; labels show measured feasibility, while the right panel highlights the strongest tested settings.



3x3 · $p=3$

PF 11.86%

Popt 4.15% · best score 2.70

4x4 · $p=2$

Optimum 3.08

Only tested depth with zero gap

4x2 · $p=3$

PF 43.99%

Popt 6.11% · best score 3.25

Functional $\neq$ monotonic with depth



GM-QAOA Works as an Ideal Feasible-Subspace Ansatz

The strongest verified result of Phase 1: feasibility and state-preparation fidelity are exactly preserved in ideal simulation.

reference_3×3

Popt 69.2%

PF = 100%
F(US) = 1.0
Circuit fidelity = 1.0
Best score = 2.70
Ideal depth = 1,062

one_to_one_4×4

Popt 10.7%

PF = 100%
F(US) = 1.0
Circuit fidelity = 1.0
Best score = 3.08
Ideal depth = 20,252

exact_capacity_4×2

Popt 50.4%

PF = 100%
F(US) = 1.0
Circuit fidelity = 1.0
Best score = 3.25
Ideal depth = 395

WHAT IS PROVED

The implemented preparation + Grover mixer preserve the fixed feasible subspaces exactly.

SCOPE

Exact fixed-size constructions—not a general scalable library.



The End-to-End Pipeline Reached Real Hardware

A controlled comparison: same backend, same job, same shots, same pass manager, same hardware options.

ibm_marrakesh · p=1 · 1,024 shots per circuit · optimization level 3 · dynamical decoupling · gate twirling

STANDARD QAOA

3.25

best feasible score
exact optimum · gap 0.00

Observed probability

PF 7.71%

Popt 1.56%

Physical resources

Depth 143

CZ 63

GM-QAOA

3.25

best feasible score
exact optimum · gap 0.00

Observed probability

PF 11.91%

Popt 2.83%

Physical resources

Depth 1,357

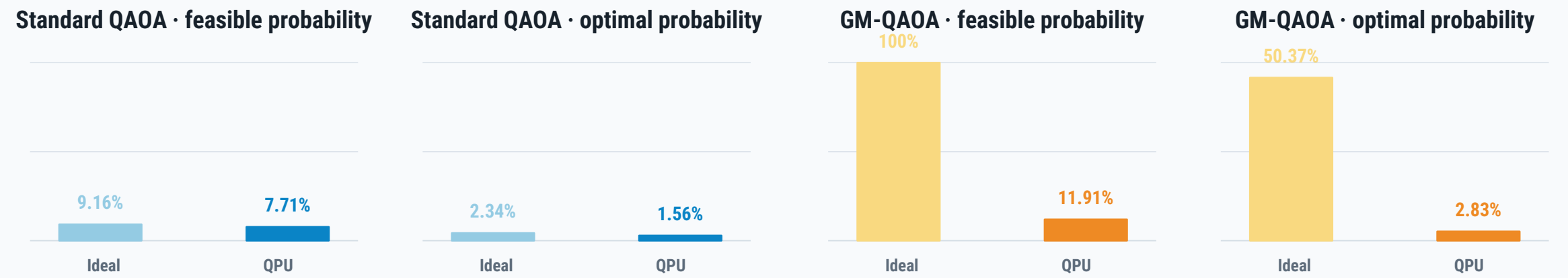
CZ 496

Both hardware distributions contained the exact optimum bitstring 10011001 · Runtime job d9vl8qfo3ppc73akfak0



Best-Sample Success Is Real—but It Is Not Statistical Reliability

The optimum appeared in both distributions, while ideal-to-hardware concentration degraded—especially for GM-QAOA.



STANDARD QAOA

16 / 1,024 optimal-state counts

Its ideal optimum probability lies inside the physical Wilson interval.

GM-QAOA

29 / 1,024 optimal-state counts

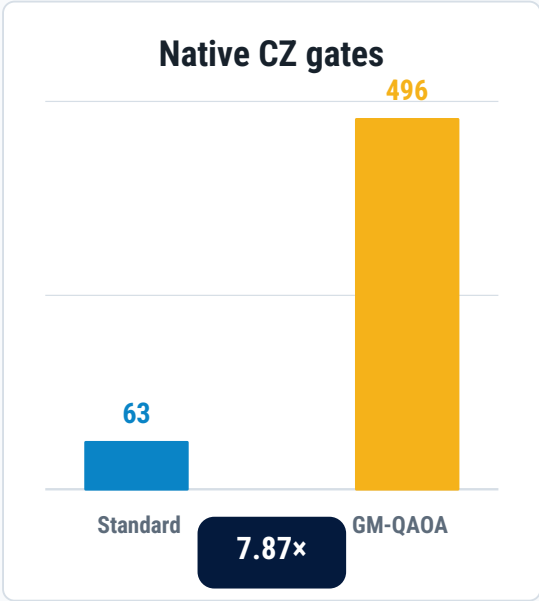
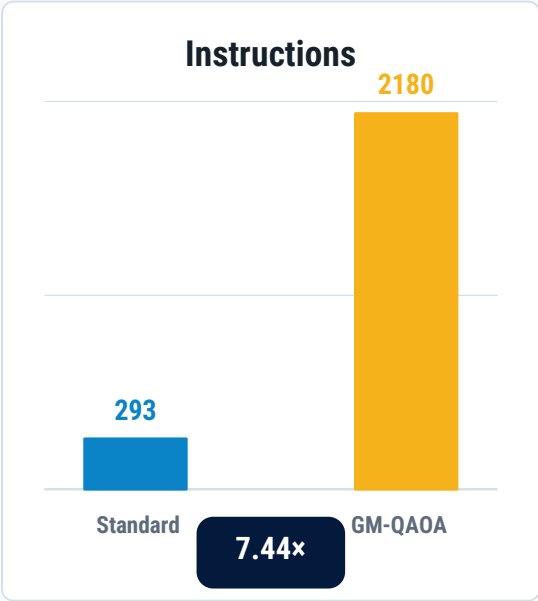
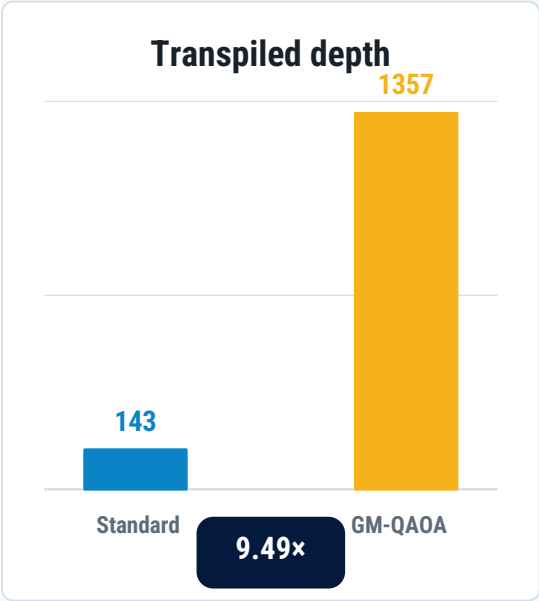
Its ideal 50.37% optimum probability is far outside the physical interval.

Correct interpretation: “the optimum was sampled” — not “the optimum was found with high probability.”



The Ideal GM-QAOA Advantage Collapses Under Physical Overhead

The data strongly support circuit length and synthesis/routing cost as major explanatory factors—without proving depth alone is causal.



GLOBAL DISTRIBUTION ERROR		
TVD	0.208	0.881
Hellinger	0.235	0.828
Classical fidelity	0.893	0.098
	Standard	GM-QAOA

PRIMARY SCIENTIFIC INTERPRETATION

GM-QAOA does not fail as an ideal ansatz. The specific fixed state preparation and selective-phase implementation are too costly to preserve the ideal advantage on this hardware execution.



A Combination-Level Novelty Result—Stated Cautiously

The individual ingredients have precedents; the searched corpus did not reveal the same controlled experimental combination.

THE EXPERIMENTAL INTERSECTION



“To the extent of the literature search conducted, this work presents an experimental combination not previously reported.”

Narrow claim: novelty lies in the controlled combination—not in inventing QAOA, GM-QAOA, QUBO assignment, IBM hardware execution, or the individual metrics.

CLOSEST PRECEDENTS

- Rhodes et al. (2025) — constrained vs. penalty mixer on IBM hardware
- Bärttschi & Eidenbenz (2020) — GM-QAOA method
- Stollenwerk et al. (2020) — hard-constrained assignment / resource estimates
- Golden et al. (2023) — Grover vs. transverse mixer scaling
- Hadfield et al. (2019) — alternating-operator framework

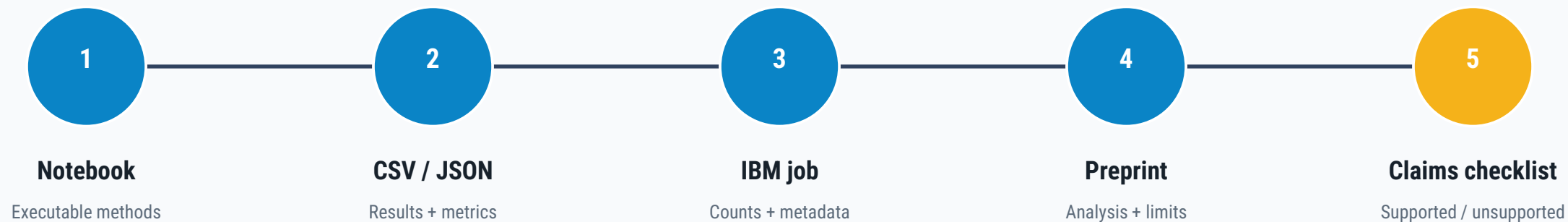
SEARCH CAVEAT

36 full-text articles were reviewed, but coverage was heterogeneous and not exhaustive of arXiv, IEEE Xplore, or Scopus. The conclusion remains provisional.



Evidence Over Hype: Every Claim Has an Artifact Boundary

This evidence-first framing aligns the presentation with QWorld's scientific-integrity principles.



SUPPORTED BY THE ARTIFACT

- One-to-one and fixed exact-capacity QUBOs are implemented and validated.
- Standard QAOA is compared at $p=1,2,3$ in ideal simulation.
- Fixed GM-QAOA circuits preserve the feasible subspace ideally.
- Both physical circuits sampled the exact optimum.
- The recorded GM-QAOA circuit has much higher depth and CZ count.

NOT SUPPORTED BY THE ARTIFACT

- Quantum advantage or practical runtime superiority.
- High-probability optimum recovery on hardware.
- QPU preservation of the GM-QAOA feasible subspace.
- A general scalable state-preparation construction.
- Universal overhead ratios across devices or instances.



Phase-1 Conclusions and the Next Research Decisions

A balanced summary: verified achievements first, then the experiments required to generalize them.

1 GM-QAOA works ideally

Exact fixed-instance preparations preserve feasibility and produce much higher $p=1$ optimum probabilities.

2 Standard QAOA works

It recovers optimal scores in multiple settings; depth effects are instance- and metric-dependent.

3 The hardware pipeline works

Both matched $p=1$ circuits executed on `ibm_marrakesh` and sampled the optimum.

4 The physical trade-off is measurable

GM-QAOA's state-preparation/mixer cost destroys most of its ideal distribution advantage.

5 The combination appears underreported

No exact methodological match was found in the searched corpus; this remains a cautious, provisional claim.

NEXT RESEARCH DECISIONS

01 Repeat matched jobs across time and multiple backends.

02 Replace generic fixed preparation with structured scalable circuits.

03 Run mitigation and ablation controls for DD, twirling, readout, and postselection.

04 Construct controlled size/capacity families and report total resource cost.

NO QUANTUM-ADVANTAGE CLAIM

The contribution is a reproducible formulation, a controlled hardware experiment, and evidence-based error/resource analysis.

THE TAKEAWAY

**The feasible-subspace construction works.
Standard QAOA works.
The complete pipeline reaches real hardware.
The evidence is reproducible and bounded.**

Within the searched corpus, the controlled experimental combination was not previously reported.

No quantum advantage is claimed. Phase 1 establishes a transparent foundation for the next experimental campaign.

PHASE 1

COMPLETE

QI26-20
QUBO · QAOA · GM-QAOA
IBM QUANTUM

QUESTIONS